

Week 10 — Forecasting

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The big picture. The best predictor is the conditional expectation; for causal ARMA we forecast via the ψ -weights (future innovations = 0) and the mean-square error grows with the horizon.

The whole point of fitting a model is to predict the future. We want X_{n+h} from the observed data $X_1, \dots, X_n$: $h > 0$ is the **horizon** (or **lead**), n is the **origin**. Throughout we pretend the model *and* its parameters are **known**, so the only randomness left is that of the future innovations. The recurring message: optimal prediction needs nothing but the covariance structure $\{\gamma_\tau\}$, and for a causal ARMA it is computed by writing the future and *erasing the future innovations*.

Key definitions

Definition — Prediction mean square error (PMSE)

For any predictor $g(X_1, \dots, X_n)$ of X_{n+h} , its quality is

$$P_{n+h}^n := \mathbb{E} \left[(X_{n+h} - g(X_1, \dots, X_n))^2 \right].$$

Smaller is better; minimising it over a class of predictors defines the “best” one in that class.

Definition — Best linear predictor (BLP)

The best *linear* predictor is the linear combination

$$X_{n+h}^n = P_{X_1, \dots, X_n}(X_{n+h}) = \sum_{j=1}^n \beta_j X_j$$

minimising the PMSE. Here $P_{X_1, \dots, X_n}(\cdot)$ is the orthogonal projection onto the span of the data. For a **Gaussian** series the conditional expectation is already linear, so the BLP *is* the overall best predictor; in general $\mathbb{E}[X_{n+h} \mid \cdot]$ is nonlinear and we restrict to linear predictors, which need only second-order information.

Definition — h -step forecast error

$e_n(h) := X_{n+h} - X_{n+h}^n$. It is unbiased, $\mathbb{E}[e_n(h)] = 0$, and its variance is exactly the PMSE: $\text{Var}(e_n(h)) = P_{n+h}^n$. For a causal ARMA, $e_n(h) = \sum_{j=0}^{h-1} \psi_j \varepsilon_{n+h-j}$.

Definition — Three sources of uncertainty

A forecast carries (1) **model** uncertainty (wrong model form), (2) **estimation** uncertainty (parameters estimated, not known), (3) **innovation** uncertainty (random future ε 's). We assume the model is known and parameters are exact, killing (1) & (2) and leaving only innovation uncertainty. (Bootstrap can restore the others; not covered.)

Definition — Gaussian prediction interval

For a Gaussian process $e_n(h) \sim \mathcal{N}(0, \text{Var}(e_n(h)))$, so a $(1 - \alpha)$ interval for X_{n+h} is

$$X_{n+h}^n \pm z_{1-\alpha/2} \sqrt{\text{Var}(e_n(h))}, \quad z_{0.975} \approx 1.96.$$

Key results & formulas

Lemma — the conditional expectation is optimal

$g = \mathbb{E}[X_{n+h} | X_1, \dots, X_n]$ minimises the PMSE over *all* predictors.

Idea: for any Y , $\mathbb{E}[(Y - c)^2] = \text{Var}(Y) + (\mathbb{E}[Y] - c)^2$ is minimised at $c = \mathbb{E}[Y]$; apply this conditionally and take the tower expectation. This is the gold standard; for Gaussian data it is linear, motivating the BLP in general.

Theorem — prediction equations (zero-mean stationary)

The BLP coefficients β solve the orthogonality conditions $\mathbb{E}[(X_{n+h} - X_{n+h}^n)X_k] = 0$ for $k = 1, \dots, n$, i.e. *the error is uncorrelated with every observation*. In matrix form $\Gamma_n \beta = \gamma_{[h]}$ with $\Gamma_n = [\gamma_{i-j}]$ (Toeplitz) and $\gamma_{[h]} = (\gamma_{n+h-1}, \dots, \gamma_h)^\top$. If Γ_n is invertible,

$$X_{n+h}^n = \gamma_{[h]}^\top \Gamma_n^{-1} X, \quad P_{n+h}^n = \gamma_0 - \gamma_{[h]}^\top \Gamma_n^{-1} \gamma_{[h]}.$$

Valid for *any* stationary process; the BLP is always unique even if Γ_n is singular. With mean μ , just centre: $X_{n+h}^n = \mu + \gamma_{[h]}^\top \Gamma_n^{-1} (X - \mu \mathbf{1}_n)$, same PMSE.

Theorem — causal ARMA forecast via ψ -weights

For a causal ARMA, $X_t = \sum_{j \geq 0} \psi_j \varepsilon_{t-j}$, the BLP and its PMSE are

$$X_{n+h}^n = \sum_{j=h}^{\infty} \psi_j \varepsilon_{n+h-j} = \psi_h \varepsilon_n + \psi_{h+1} \varepsilon_{n-1} + \dots, \quad P_{n+h}^n = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2.$$

Idea: every past-based predictor is a combination of ε_s , $s \leq n$; matching it to the observable part of X_{n+h} kills all controllable error, leaving the h future innovations. **Practical rule:** write $X_{n+h} = \sum_j \psi_j \varepsilon_{n+h-j}$ and set $\varepsilon_{n+1}, \dots, \varepsilon_{n+h} = 0$.

Proposition — limiting behaviour

For causal ARMA, $\psi_j \rightarrow 0$ exponentially, so as $h \rightarrow \infty$:

$$X_{n+h}^n \rightarrow \mu, \quad \text{Var}(e_n(h)) = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2 \rightarrow \sigma^2 \sum_{j \geq 0} \psi_j^2 = \gamma_0.$$

The far future is just the unconditional distribution: forecast = mean, variance = marginal variance.

Proposition — integrated (ARIMA) forecasts

For ARIMA(p, d, q), $Z_t = \nabla^d X_t$ is ARMA(p, q); forecast Z , then *re-integrate*: $\nabla^d X_{n+h}^n = Z_{n+h}^n$. For $d = 1$,

$$X_{n+h}^n = X_n + \sum_{j=0}^{h-1} Z_{n+h-j}^n.$$

If $\mathbb{E}[Z_t] = \alpha \neq 0$ then $X_{n+h}^n \approx X_n + \alpha h$ (a drift line). Crucially the PMSE **diverges**, $P_{n+h}^n \rightarrow \infty$: no mean reversion.

Truncated recipe (the hand method, ARMA & ARIMA). (1) Isolate y_t on the left, everything else on the right. (2) Replace $t \rightarrow n + h$. (3) On the right substitute: future obs $\rightarrow$ their forecasts, future errors $\rightarrow 0$, past errors $\rightarrow$ residuals $\hat{\varepsilon}_s$ (pre-sample $\rightarrow 0$). Run $h = 1, 2, \dots$ in order; each forecast feeds the next.

Intuition

Why the interval widens, then saturates

A point forecast is useless without its spread. The error $e_n(h)$ accumulates the unknown $\varepsilon_{n+1}, \dots, \varepsilon_{n+h}$, so the interval grows with h . For *stationary* ARMA the growth saturates: as $h \rightarrow \infty$, $\text{Var}(e_n(h)) \rightarrow \gamma_0$, so the band converges to “mean $\pm$ a few marginal standard deviations” — forecasting far ahead beats nothing. For *integrated* (ARIMA) processes it grows without bound.

“Write the future, erase the future innovations”

The two ARMA theorems are one projection seen two ways. Prediction equations project onto $\text{span}\{X_1, \dots, X_n\}$ using the ACVS; the ψ -weight formula projects onto the *innovation* space $\text{span}\{\varepsilon_s : s \leq n\}$. For invertible ARMA these spans coincide, so answers agree — but the innovation view is far easier: take the $\text{MA}(\infty)$ form and drop every ε you have not yet seen.

Mean reversion vs. memory length

Stationary models forget: an $\text{AR}(1)$ gives $X_{n+h}^n = \phi^h X_n \rightarrow 0$, decaying smoothly to the mean. An $\text{MA}(q)$ has memory only q : for $h > q$ the forecast jumps straight to μ . Integrated models never forget — a random walk with drift forecasts the drift line $X_n + \alpha h$ forever.

AR(p) shortcut

For an $\text{AR}(p)$ with $n \geq p$, the one-step forecast is simply $X_{n+1}^n = \sum_{i=1}^p \phi_i X_{n+1-i}$ — no matrix to invert. Reason: subtracting the AR recursion leaves ε_{n+1} , which is uncorrelated with the past, so the prediction equations are satisfied automatically.

MA(2) one- to three-step forecast

Let $X_t = \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2}$, $\varepsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$. Here $\psi_0 = 1, \psi_1 = -\theta_1, \psi_2 = -\theta_2, \psi_j = 0$ for $j \geq 3$. Zeroing future innovations:

$$X_{n+1}^n = -\theta_1 \hat{\varepsilon}_n - \theta_2 \hat{\varepsilon}_{n-1}, \quad X_{n+2}^n = -\theta_2 \hat{\varepsilon}_n, \quad X_{n+3}^n = 0 (= \mu).$$

Variances $\text{Var}(e_n(h)) = \sigma^2, \sigma^2(1 + \theta_1^2), \sigma^2(1 + \theta_1^2 + \theta_2^2) = \gamma_0$. A 95% interval for X_{n+2} : $-\theta_2 \hat{\varepsilon}_n \pm 1.96 \sigma \sqrt{1 + \theta_1^2}$.

Key takeaways

Key takeaways — the week’s reflexes

- **Best predictor** = $\mathbb{E}[X_{n+h} \mid \text{data}]$; for Gaussian data this is linear, so it equals the BLP. Linear prediction needs only the ACVS.
- **Solve** $\Gamma_n \beta = \gamma_{[h]}$ for the general BLP; centre by μ if the mean is nonzero. The error is orthogonal to every observation.
- **Causal ARMA**: write $X_{n+h} = \sum_j \psi_j \varepsilon_{n+h-j}$ and set future $\varepsilon \rightarrow 0$. $\text{PMSE} = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2$; get ψ -weights from $\Theta(B)/\Phi(B)$.
- **Limits**: stationary $\Rightarrow X_{n+h}^n \rightarrow \mu$ and $\text{Var}(e_n(h)) \rightarrow \gamma_0$ (band saturates). ARIMA $\Rightarrow$ drift line and $\text{PMSE} \rightarrow \infty$ (band fans out).
- **ARIMA**: difference $\rightarrow$ forecast the ARMA $\rightarrow$ re-integrate; or apply the truncated recipe to the expanded unit-root equation directly.
- **Interval**: $X_{n+h}^n \pm 1.96 \sqrt{\text{Var}(e_n(h))}$ for 95% (Gaussian).